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Quick Introduction to Linear Algebra

JOHNS HOPKINS
UNIVERSITY

Basics

- Consider one or more equations of the form $a_1x_1 + a_2x_2 \ldots a_nx_n=b$
- Let's say you want to solve for values for x
- Construct a matrix:

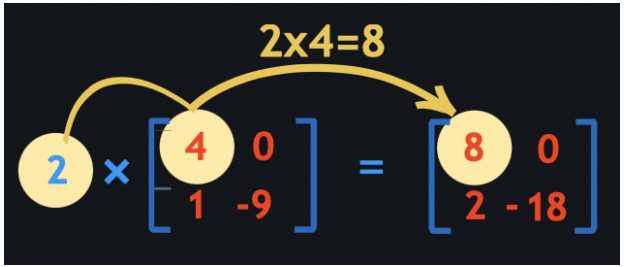
$$3x + 4y + 7z = 4$$
$$2x + y - 3z = 7$$
$$5x + 9y + 8z = -1$$

$$A=\begin{bmatrix} 3 & 4 & 7 \\ 2 & 1 & -3 \\ 5 & 9 & 8 \end{bmatrix} x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} b = \begin{bmatrix} 4 \\ 7 \\ -1 \end{bmatrix}$$

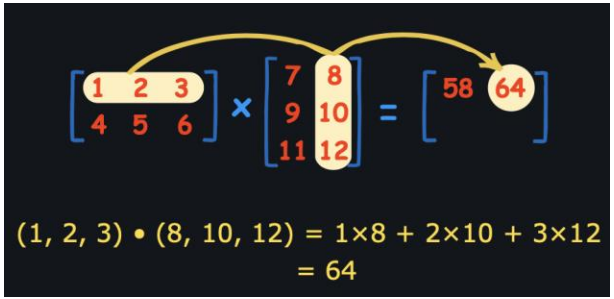
- Each row of a matrix is a vector and the matrix is the plane that contains these vectors, note that other vectors exist in the same plane

Matrix Operations

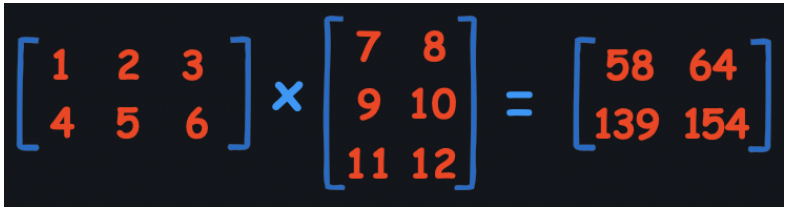
- Scalar Multiplication



- Dot product multiplication



[1]



Matrix Determinant

- This can only be calculated for square matrices
- Matrix Determinant Calculation

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = 1*4-2*3=4-6=-2$$

- This calculation gets more complex the higher the dimensions, so maybe we want to find something to do this calculation.
- The determinant can be used to determine the size of a triangle (3x3 matrix with 1's for the 3rd column), if the determinant is 0 the three points are collinear and so much more fun stuff.

[1]

References

- [1] 2021. mathisfun.com. How to Multiply Matrices. Found at https://www.mathsisfun.com/algebra/matrix-multiplying.html.

JOHNS HOPKINS
UNIVERSITY

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